

● EASY

● ALGEBRA

● ANSWER KEY

Linear equations in one variable

02

10 answers · Rationale-rich · Teach from doc

Q1**40**

The correct answer is 40. Subtracting 5 from both sides of the given equation yields $4x = 160$. Dividing both sides of this equation by 4 yields $x = 40$. Therefore, the solution to the given equation is 40.

Q2**27**

The correct answer is 27. Multiplying both sides of the given equation by 3 yields $36 + x = 39$, or $18 + 3x = 27$. Therefore, the value of $18 + 3x$ is 27.

Q3**A**

A. 5

Choice A is correct. It's given that a principal used a total of 25 blue flags and yellow flags. It's also given that of the 25 flags used, 20 flags were blue. Subtracting the number of blue flags used from the total number of flags used results in the number of yellow flags used. It follows that the number of yellow flags used is $25 - 20$, or 5.

Choice B is incorrect. This is the number of blue flags used.

Choice C is incorrect. This is the total number of flags used.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4**B****B.** 18

Choice B is correct. Subtracting 12 from each side of the given equation yields $3x - 12 = 30 - 12$, or $3x - 12 = 18$. Therefore, the value of $3x - 12$ is 18.

Choice A is incorrect. This is the value of $x - 12$, not $3x - 12$.

Choice C is incorrect. This is the value of $x + 12$, not $3x - 12$.

Choice D is incorrect. This is the value of $3x + 12$, not $3x - 12$.

Q5**8.6**

Accept also: 43/5

The correct answer is 8.6. Dividing both sides of the given equation by 10 yields $x = 8.6$. Therefore, 8.6 is the solution to the given equation. Note that 8.6 and 43/5 are examples of ways to enter a correct answer.

Q6

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Answer not provided in source data — see rationale below.

Choice B is correct. Subtracting the number of books left over from the total number of books results in $43 - 7 = 36$, which is the number of books distributed. Dividing the number of books distributed by the number of books given to each child results in $\frac{36}{2} = 18$.

Choice A is incorrect and results from dividing the total number of books by the number of books given to each child, $\frac{43}{2} \approx 22$, then subtracting the number of books left over from the result, $22 - 7 = 15$. Choice C is incorrect and results from adding the number of books left over to the total number of books, $43 + 7 = 50$, then dividing the result by the number of books given to each child, $\frac{50}{2} = 25$. Choice D is incorrect and results from dividing the total number of books by the number of books given to each child, $\frac{43}{2} \approx 22$, then adding the number of books left over, $22 + 7 = 29$.

Q7

c

C. 54

Choice C is correct. It's given that $8x = 6$. Multiplying each side of this equation by 9 yields $72x = 54$. Therefore, the value of $72x$ is 54.

Choice A is incorrect. This is the value of $4x$, not $72x$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8**D**

D. $8x + 3 = 83$

Choice D is correct. The given phrase “8 times a number x ” can be represented by the expression $8x$. The given phrase “3 more than” indicates an increase of 3 to a quantity. Therefore “3 more than 8 times a number x ” can be represented by the expression $8x + 3$. Since it’s given that 3 more than 8 times a number x is equal to 83, it follows that $8x + 3$ is equal to 83, or $8x + 3 = 83$. Therefore, the equation that represents this situation is $8x + 3 = 83$.

Choice A is incorrect. This equation represents 3 times the quantity 8 times a number x is equal to 83.

Choice B is incorrect. This equation represents 8 times a number x is equal to 3 more than 83.

Choice C is incorrect. This equation represents 8 more than 3 times a number x is equal to 83.

Q9**480**

The correct answer is 480. Multiplying both sides of the given equation by 8 yields $82 + x = 860$, or $16 + 8x = 480$. Therefore, if $2 + x = 60$, the value of $16 + 8x$ is 480.

Q10**24**

The correct answer is 24. Multiplying both sides of the given equation by 3 yields $3(3x + 2) = 24$. Using the distributive property to rewrite the left-hand side of this equation yields $9x + 6 = 24$.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Answers recovered from rationale for Q6 — the source CB data omitted a structured key for these items, so the answer slot shows —. The worked solution in the rationale below each entry contains the correct value.

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